Clearance & Elimination Practice 2

Module 4 - Drug Elimination and Clearance · Daily practice sheet · No date

Name _____ Date _____

1a 1. A single 250-mg oral dose of an antibiotic is given to a young man (age 32 years, creatinine clearance 122 mL/min, 78 kg). From the literature, the drug is known to have an apparent V_D equal to 21% of body weight and an elimination half-life of 2 hours. The dose is normally 90% bioavailable. Urinary excretion of the unchanged drug is equal to 70% of the absorbed dose. What is the total body clearance for this drug? (Give it in L/hr.)

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1b 1. A single 250-mg oral dose of an antibiotic is given to a young man (age 32 years, creatinine clearance 122 mL/min, 78 kg). From the literature, the drug is known to have an apparent V_D equal to 21% of body weight and an elimination half-life of 2 hours. The dose is normally 90% bioavailable. Urinary excretion of the unchanged drug is equal to 70% of the absorbed dose. What is the renal clearance for this drug?

1c 1. A single 250-mg oral dose of an antibiotic is given to a young man (age 32 years, creatinine clearance 122 mL/min, 78 kg). From the literature, the drug is known to have an apparent V_D equal to 21% of body weight and an elimination half-life of 2 hours. The dose is normally 90% bioavailable. Urinary excretion of the unchanged drug is equal to 70% of the absorbed dose. What is the probable mechanism for renal clearance of this drug?

A. Filtration only

B. Cleared mainly by the liver

C. Filtration with tubular reabsorption

D. Filtration with active secretion

Problem 2 Estimate the creatinine clearance of a 37-year-old female who weighs 143 lb, serum creatinine of 0.6 mg/dL, and stands 5’6” tall.

Solutions

Her key: the answer and working on her own solved sheet or annotated slide. Worked by the drill: the drill's own working, where her sheet gives no worked value.

1a Her key

Answer: 5.676 L/hr

WHAT THE STEM GIVES

D_0	250-mg	*not needed: Cl_T depends on V_D and k alone*
age	32 years	*not needed*
CrCl	122 mL/min	*not needed for the number*
weight	78 kg	$V_D = 0.21 \times 78 = 16.38$ L
V_D	21% of body weight	0.21 L per kg, times 78 kg, multiplies k
t½	2 hours	gives $k = \frac{0.693}{2} = 0.3465\ hr^{-1}$
F	90% bioavailable	*not needed*
f_e	70% of the absorbed dose	*not needed: the total, not the renal part, is asked*

THE WORKING, ONE LINE AT A TIME

UNIT

$$V_D = (0.21)(78\ kg) = 16.38\ L$$

A volume given as a percentage of body weight is read as litres per kilogram: 21% of 78 kg is 0.21 × 78 = 16.38 L.

SETUP

$$Cl_T = V_D \times k,\ k = \frac{0.693}{2\ hr}$$

Total clearance is the elimination rate constant times the volume of distribution: reciprocal hours times litres gives L/hr.

ALGEBRA

$$Cl_T = (16.38\ L)(0.3465\ hr^{-1}) = 5.676\ L/hr$$

$k = \frac{0.693}{2\ hr} = 0.3465\ hr^{-1}$; $(16.38\ L)(0.3465\ hr^{-1}) = 5.676$ L/hr, litres times reciprocal hours giving litres per hour. Total clearance Cl_T depends on V_D and k alone, so the 250 mg dose, the 90% bioavailability and the 70% urinary share are not used here.

> **DOES THE NUMBER MAKE SENSE?**
> With a 2-hour half-life about a third of the 16.38 L is cleared each hour ($k = 0.3465\ hr^{-1}$), so Cl_T is about 5.676 L/hr, a little over a third of 16.38.

1a Her key

Answer: 94.59 mL/min

WHAT THE STEM GIVES

D_0	250-mg	*not needed: Cl_T depends on V_D and k alone*
age	32 years	*not needed*
CrCl	122 mL/min	*not needed for the number; a comparison for later*
weight	78 kg	$V_D = 0.21 \times 78 = 16.38$ L
V_D	21% of body weight	0.21 L per kg, times 78 kg, multiplies k
t½	2 hours	gives $k = \frac{0.693}{2} = 0.3465\ hr^{-1}$, so Cl_T = 5.676 L/hr in part a
F	90% bioavailable	*not needed*
f_e	70% of the absorbed dose	*not needed: the total is asked*

THE WORKING, ONE LINE AT A TIME

UNIT

$$(5.676\ \text{L/hr})\left(\frac{1000\ \text{mL}}{1\ \text{L}}\right)\left(\frac{1\ \text{hr}}{60\ \text{min}}\right) = 94.59\ \text{mL/min}$$

Litres become millilitres (× 1000) and hours become minutes (÷ 60), so total clearance is in the same unit as CrCl.

> **DOES THE NUMBER MAKE SENSE?**
> One L/hr is 1000 mL in 60 min, about 17 mL/min; 5.676 L/hr is a little under six of those, so a value near 95 mL/min fits, and 94.59 mL/min does.

1b

Her key

Answer: 66.22 mL/min

WHAT THE STEM GIVES

D_0	250-mg	*not needed*
age	32 years	*not needed*
CrCl	122 mL/min	*not needed for the number; Cl_R is compared with it afterwards*
weight	78 kg	$V_D = 0.21 \times 78 = 16.38$ L
V_D	21% of body weight	with k gave Cl_T = 5.676 L/hr = 94.59 mL/min in part a
t½	2 hours	gives $k = \frac{0.693}{2} = 0.3465\ hr^{-1}$
F	90% bioavailable	*not needed*
f_e	70% of the absorbed dose	f_e = 0.7 multiplies Cl_T

THE WORKING, ONE LINE AT A TIME

SETUP

$Cl_R = f_e \times Cl_T$, $f_e = 0.7$

Urinary excretion of unchanged drug is 70% of the absorbed dose, so $f_e = 0.7$ and renal clearance is 0.7 of the total.

ALGEBRA

$Cl_R = (0.7)(94.59\ mL/min) = 66.22\ mL/min$

Using the unrounded total clearance, 94.594 mL/min, gives 66.22; the rounded 94.59 gives 66.21. Both are the same answer.

> **DOES THE NUMBER MAKE SENSE?**
> Seven tenths of 94.59 mL/min is a little under 70: 66.22 mL/min. Cl_R cannot exceed the total clearance, 94.59 mL/min, and it is below the CrCl of 122 mL/min, so filtration with reabsorption.

0.7 × 94.59 gives 66.21; her key prints 66.22 because it carries the unrounded 94.594. Either is accepted.

1c

Her key

Answer: C. Filtration with tubular reabsorption

WHAT THE STEM GIVES

D_0	250-mg	*not needed: clearance does not depend on the dose*
age	32 years	*not needed: the creatinine clearance is given, so no Cockcroft-Gault*
CrCl	122 mL/min	the GFR estimate that Cl_R is compared with
weight	78 kg	scales the volume: (0.21)(78) = 16.38 L
V_D	21% of body weight	0.21 L per kg, inside $Cl_T = kV_D$
t½	2 hours	gives $k = \frac{0.693}{2} = 0.3465\ hr^{-1}$
F	90%	*not needed: f_e is already a fraction of the absorbed dose*
f_e	70% of the absorbed dose	$f_e = 0.70$, scales Cl_T to Cl_R

THE WORKING, ONE LINE AT A TIME

SETUP

$f_e = 0.70$: urinary excretion of unchanged drug is 70% of the absorbed dose

f_e is the fraction of the absorbed dose reaching the urine unchanged. It says the kidney handles 70% of the elimination, not how: filtration alone, with reabsorption, or with secretion.

SETUP

$V_D = (0.21)(78\ kg) = 16.38\ L$; $k = \frac{0.693}{2\ hr} = 0.3465\ hr^{-1}$; $Cl_T = kV_D = (0.3465)(16.38) = 5.676\ L/hr$

The mechanism is read from renal clearance, a share of total clearance, so total clearance comes first: the volume from 21% of body weight and k from the 2-hour half-life.

UNIT

$$Cl_T = 5.676 \text{ L/hr} \times \frac{1000 \text{ mL}}{1 \text{ L}} \times \frac{1 \text{ hr}}{60 \text{ min}} = 94.59 \text{ mL/min}$$

The creatinine clearance, the GFR estimate, is in mL/min, so total clearance is put in the same unit before the renal share is taken.

ALGEBRA

$Cl_R = f_e \times Cl_T = 0.70 \times 94.59 \text{ mL/min} = 66.22 \text{ mL/min}$

Renal clearance is the renal fraction of total clearance, a number with units that can be set against the filtration rate. The unrounded 94.594 gives 66.22; the rounded total gives 66.21.

SETUP

Compare: $Cl_R = 66.22$ mL/min against the GFR, the creatinine clearance of 122 mL/min (about 120); 66.22 is below it

A renal clearance below the GFR means some filtered drug is taken back into the blood: filtration with partial reabsorption. Above the GFR would mean active secretion.

> **DOES THE NUMBER MAKE SENSE?**
> $f_e = 0.70$ alone cannot name the mechanism: a drug with the same f_e but a much shorter half-life would have a renal clearance above the GFR and be secreted. Only Cl_R in mL/min set against the GFR, 122 mL/min for this patient, decides, and 66.22 is below it.

Renal clearance, 66.22 mL/min, is below the creatinine clearance, 122 mL/min, this patient's GFR estimate. Filtration alone would clear plasma at that rate, so some filtered drug moved back into the blood: filtration with partial reabsorption.

Problem 2 Her key

Answer: 120.2 mL/min

WHAT THE STEM GIVES

age	37-year-old	140 − 37 = 103 on top
sex	female	IBW starts at 45.5 kg and the result is multiplied by 0.85
weight	143 lb	*not needed: her keys use ideal body weight*
S_{Cr}	0.6 mg/dL	the denominator, 72 × 0.6 = 43.2
height	5’6”	6 inches over 5 ft, so IBW = 45.5 + 2.3 × 6 = 59.3 kg

THE WORKING, ONE LINE AT A TIME

UNIT

5 ft 6 in: 6 inches over 5 ft

The ideal body weight equation counts only the inches above 5 ft (60 in). A height of 5 ft 6 in is already in feet and inches, so no conversion from centimetres is needed and the inches over 5 ft are the 6 in read straight off the height.

ALGEBRA

$IBW = 45.5 + 2.3(6) = 59.3$ kg

The female form starts at 45.5 kg. The 143 lb total body weight is not used, because her keys use ideal body weight.

SETUP

$$\text{CrCl} = 0.85 \times \frac{(140 - 37)(59.3)}{(72)(0.6)}$$

The Cockcroft-Gault equation estimates creatinine clearance from age, ideal body weight and serum creatinine S_{Cr}: (140 − age)(IBW) over 72 × S_{Cr}. The patient is female, and women produce less creatinine per kilogram of body weight, so the male estimate is multiplied by 0.85.

ROUND

$$\text{CrCl} = 120.2 \text{ mL/min}$$

(103)(59.3) = 6107.9 on top and (72)(0.6) = 43.2 below; $\frac{6107.9}{43.2}$ = 141.4, and 0.85 × 141.4 = 120.2 mL/min. CrCl is reported in mL/min, the unit of the glomerular filtration rate it estimates, to one decimal place as her key does.

DOES THE NUMBER MAKE SENSE?

A 37-year-old woman with a low S_{Cr} of 0.6 mg/dL has normal kidneys, so a CrCl near or above the usual 120 mL/min fits; 120.2 mL/min does. Forgetting the 0.85 gives a value about a sixth too high.